

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMC501 - HEAT TRANSFER

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question) 12

- 1 With increase in temperature, thermal Conductivity of metals
 - a. Increases
 - b. Decreases
 - c. is constant
 - d. is unaffected
- 2 Heat transfer rate is proportional to Fourth power of absolute temperature in
 - a. Fourier's law of conduction
 - b. Newton's law of cooling
 - c. Stefan Boltzmann Law of radiation
 - d. Kirchhoff's law
- 3 With increase in heat transfer coefficient, effectiveness of fin
 - a. Increases
 - b. Decreases
 - c. is constant
 - d. is unaffected
- 4 In case of fins, temperature distribution along its length is
 - a. Exponential
 - b. Linear
 - c. Parabolic
 - d. constant
- 5 In heat exchanger, NTU stands for
 - a. Net Transfer Units
 - b. Number of Transfer Units
 - c. New Thermal Units
 - d. None of the above
- 6 Absorptivity of a white surface is equal to
 - a. Infinite
 - b. One
 - c. Zero
 - d. Two

- 7 Which number is used to characterize the Natural Convection..?
 - a. Prandtl Number
 - b. Reynolds number
 - c. Grashof number
 - d. Biot number
- 8 With increase in temperature, thermal Conductivity of non-metals
 - a. Increases
 - b. Decreases
 - c. is constant
 - d. is unaffected
- 9 Transmissivity of a black surface is equal to
 - a. Infinite
 - b. One
 - c. Zero
 - d. Two
- 10 The exit temperature of cold fluid can be higher than exit temperature of hot fluid in the case of
 - a. Parallel flow heat exchanger
 - b. Counter flow heat exchanger
 - c. Evaporator
 - d. Condenser
- 11 Shape factor of a convex surface with respect to itself is equal to
 - a. zero
 - b. one
 - c. two
 - d. infinite
- 12 Wien's displacement law is useful to calculate
 - a. Conduction heat transfer
 - b. Convection heat transfer
 - c. Wavelength corresponding to maximum emissive power at specified temperature
 - d. Maximum emissive power at specified temperature

Q2. Solve the following.

- A) Define thermal conductivity. 6
 Derive the unit of thermal conductivity using Fourier's law of conduction.
 Explain the effect of increase & decrease in temperature on thermal conductivity of
 - a. Metals
 - b. Non metals
- B) An insulating powder is densely packed in the annular space between two concentric spheres with radii 75 mm and 50 mm. The inner sphere is uniformly heated with electric power input of 30 W. Steady state temperature attained by inner sphere is 120°C and that by outer sphere is 30°C. Neglect thermal resistance offered by the sphere.
 - A. Draw analogous electrical circuit diagram.
 - B. Calculate thermal conductivity of the powder.

Q3. Solve the following.

- A) Derive the generalized heat conduction equation in Cartesian coordinates. 6
- B) A solid steel ball 5 cm in diameter and initially at 450°C is quenched in a controlled environment at 90°C with convection coefficient of 115 W/m²K. Determine the time taken by center to reach a temperature of 150°C.
 Take thermo-physical properties as $C = 420 \text{ J/kg.K}$, $\rho = 8000 \text{ kg/m}^3$, $k = 46 \text{ W/m.K}$.
Justify the equation used for temperature distribution.

Q4. Solve Any Two of the following.

A) Define, write mathematical equation and explain physical significance of following 6 quantities

1. Reynolds number
2. Grashof number

B) A hot rectangular plate 5 cm x 3 cm maintained at 200°C is exposed to still air at 30°C. 6 Calculate percentage increase in convective heat transfer rate if smaller side of the plate is held vertical than the bigger side. Neglect ITG of the thickness.

Use correlation $Nu = 0.59 (Gr.Pr)^{0.25}$

Air properties at 115°C - Density = 0.91 kg/m³, $C_p = 1.009$ kJ/kgK, $\mu = 22.65 \times 10^{-6}$ N.s/m², $k = 0.0331$ W/mK

C) Air is flowing over a flat plate 5 m long and 2.5 m wide with a velocity of 4 m/s at 15°C. 6 If $\rho = 1.208$ kg/m³ and $\nu = 1.47 \times 10^{-5}$ m²/s, calculate:

1. Length of plate over which the BL is laminar, and thickness of BL (Laminar).
2. Local & average skin friction coefficient
3. Shear stress at the location where BL ceases to be laminar

Use following equations for thickness of velocity boundary layer and drag coefficient.

$$\frac{\delta}{x} = \frac{5}{\sqrt{Re_x}}, \quad \overline{C_f} = 1.328 Re_L^{-0.5}$$

Q5. Solve Any Two of the following.

A) Derive expression for LMTD method of analysis for parallel flow heat exchanger. 6

B) Represent temperature distribution in following heat exchangers. 6

1. Parallel flow
2. Counter flow
3. Evaporator
4. Condenser

Express the mathematical relation for overall heat transfer coefficient, if in a heat exchanger, tube wall thickness is negligible & thermal resistance of tube material can be neglected.

C) A counter flow heat exchanger is employed to cool 0.55 kg/s ($C_p = 2.45$ KJ/kg°C) of oil 6 from 115°C to 40°C by the use of water. The inlet and outlet temperatures of cooling water are 15°C and 75°C respectively. Overall heat transfer coefficient is expected to be 1450 W/m² °C. Using ϵ - NTU method, calculate the following:

- i. Mass flow rate of water
- ii. Effectiveness of heat exchanger

Q6. Solve Any Two of the following.

A) Define shape factor and provide its mathematical formula. 6

Apply shape factor algebra for following cases:

- a. Flat surface
- b. Convex Surface
- c. Concave surface
- d. Two infinitely long parallel plates

B) Calculate the following for an industrial furnace and emitting radiation at 2500°C. 6 Calculate

1. Monochromatic emissive power at 1.2 μ m length.
2. Wavelength at which emission is maximum
3. Maximum emissive power
4. Total emissive power
5. Total emissive power of furnace if it is assumed as a real surface with emissivity 0.9.

- C) Two large parallel plates at temperature 1000 K and 600 K have emissivity of 0.5 and 0.8 respectively. A radiation shield having emissivity 0.1 on one side and 0.05 on the other side is placed between the plates.
1. Calculate the heat transfer rate by radiation per square meter area, with and without radiation shield.
 2. Compare the reduction heat transfer with the situation when no radiation shield is placed in between, on % basis.

***** End *****